Throughput Optimization for Ambient IoT Systems in Indoor Factory Environments

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***Abstract*—The ambient Internet of Things (A-IoT) enables passive sensing and energy harvesting from ambient radio frequency (RF) signals, providing an energy-efficient solution for indoor industrial monitoring. However, dense factory environments often present limited RF energy and stringent quality-of-service (QoS) constraints, significantly hindering network throughput. To address these issues, this paper formulates the throughput maximization as a nonconvex mixed-integer nonlinear programming (MINLP) model under energy causality and QoS constraints, and introduces a two-layer adaptive bilevel decomposition algorithm combined with branch-and-bound (ABD-BB). In the outer layer, continuous relaxation efficiently identifies promising candidate devices, while the inner layer employs adaptive branch-and-bound to refine device activation and optimize power allocation. Simulation results from a realistic indoor factory scenario demonstrate that ABD-BB achieves substantial throughput gains: 310.6% over random access and 293.9% over classical Lagrangian methods, while closing the gap with exhaustive branch-and-bound to just 3.7% and requiring only approximately 12% of its computational time. These results highlight ABD-BB as an effective and computationally efficient solution for real-time throughput optimization in industrial A-IoT networks.**

***Index Terms*—Ambient IoT, Lagrange duality, Mixed-integer nonlinear programming, Throughput optimization**

I. Introduction

The Ambient Internet of Things (A-IoT) has emerged as a promising technology in wireless communications, enabling simultaneous energy harvesting and data transmission from existing ambient radio frequency (RF) signals, thereby significantly reducing energy costs and infrastructure complexity [1]. In the era of Industry 4.0, the rapid proliferation of IoT devices in indoor factory environments has made traditional wired connections increasingly impractical due to their installation complexity and limited scalability [2].

Ambient backscatter communication (AmBC), a core component of A-IoT, achieves ultra-low power communications by passively reflecting ambient RF signals, making it highly suitable for battery-constrained and battery-free scenarios [3], [4]. However, the typically low RF energy density and strict quality of service (QoS) constraints in dense industrial environments pose substantial challenges to the maximization of performance, necessitating joint optimization of device activation, which selects the active subset of devices, and power allocation, which assigns appropriate transmission power levels to these activated devices [5].

Existing studies have explored various strategies to address energy optimization, energy efficiency, and QoS guarantees in A-IoT systems [6]. Galappaththige *et al.* utilized non-orthogonal multiple access (NOMA) to minimize power consumption in AmBC networks while ensuring QoS [7]. Research on optimization of performance fairness in LoRa networks demonstrated improved performance through joint optimization of energy harvesting duration and transmission power [8]. Although the Aloha-NOMA scheme partially resolves the trade-off between energy efficiency and network performance, further improvements are needed to achieve joint optimization of device activation and power allocation [9]. Recent advances in multi-hop communications and two-tier network optimization suggest potential performance improvements of A-IoT networks by enabling more efficient RF resource utilization and reducing interference. However, effectively integrating these techniques into A-IoT systems remains an open research problem [10].

Despite the aforementioned efforts, addressing the joint throughput maximization problem under RF energy harvesting, energy causality, and QoS constraints remains largely unresolved, especially in indoor factory environments. Such environments typically feature relatively enclosed spaces with limited RF energy sources and large-scale device deployments, creating unique challenges in promptly handling task demands and efficiently coordinating both operations and energy harvesting among multiple devices. Consequently, these problems are often formulated as nonconvex mixed-integer nonlinear programming (MINLP) models, posing substantial computational challenges for conventional optimization methods.

To address these limitations, this paper proposes an adaptive bilevel decomposition with branch-and-bound (ABD-BB) algorithm. This approach first leverages continuous relaxation to efficiently identify promising activation candidates, followed by adaptive branch-and-bound for precise optimization. Simulation results demonstrate that the ABD-BB algorithm significantly outperforms traditional methods, such as classical Lagrangian optimization and exhaustive branch-and-bound, in terms of throughput enhancement and computational efficiency.

The remainder of this paper is organized as follows. Section II introduces the system model, Section III describes the proposed ABD-BB algorithm and benchmark methods, Section IV presents simulation results, and Section V concludes the paper.

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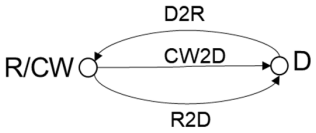


Fig. 1. D1T1-A2 Topology Deployment

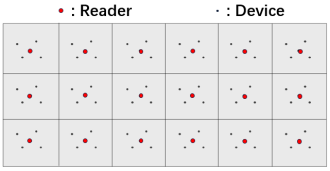


Fig. 2. Indoor Factory Model

II. System Model and Problem Description

A. Indoor Factory Model and Channel Modeling

This paper considers a typical industrial indoor environment, based on the standard D1T1-A2 topology architecture for system deployment and analysis. In this architecture (illustrated in Fig. 1), the continuous-wave (CW) transmitter co-located with the reader provides energy to devices via the CW2D link; the reader sends downlink commands via the R2D (reader-to-device) link; and the backscattered signal from each device travels along the D2R (device-to-reader) link. The CW node, deployed at the same location as the reader, continuously transmits stable RF signals to support device energy harvesting and ambient backscatter communication. The devices themselves have no active transmission capability and communicate only by reflecting continuous wave signals transmitted from the CW node. This paper focuses on the throughput optimization problem when devices are deployed in indoor factories under this typical A-IoT system architecture.

To realistically model wireless propagation characteristics in indoor factory environments, this study uses channel modeling based on the 3GPP TR 38.901 indoor factory model [11]. Specifically, the path loss formula between the devices and the reader is as follows:

$$\begin{aligned} PL(d) &= PL(d_0) + 10\,n\,\log_{10}\!\left(\frac{d}{d_0}\right) + X_\sigma, \\ PL(d_0) &= 33 + 25.5\log_{10}(d_0) + 20\log_{10}(f), \end{aligned} \tag{1}$$

where $PL(d_0)$ is the path loss at the reference distance (typically $d_0 = 1$ meter), n is the path loss exponent (typical indoor values between 2 and 3), d represents the actual distance between the device and the base station, and X_σ is shadow fading following a Gaussian distribution with zero mean and standard deviation σ.

Furthermore, because the CW transmitter and reader are co-located and devices reflect the CW signal, the effective backscatter link is a cascaded reader–device–reader channel: the signal experiences path loss twice and an additional reflection loss at the device. The channel model can be expressed as:

$$h_{n,t} = \sqrt{PL(d_n)^{-1}} \cdot g_{n,t} \tag{2}$$

Here, $g_{n,t}$ represents the small-scale fading component following a complex Gaussian distribution. This modeling approach approximates the wireless channel characteristics of actual industrial environments.

The indoor factory model adopts the structural layout recommended in TR 38.901. To facilitate the deployment of the A-IoT system, the factory interior space can be divided into multiple regular grids, with one Reader placed at the center of each area. Devices are installed at a standard height of 1.5 meters, while Readers are positioned at the space center or factory ceiling at a height of 8 meters to ensure good line-of-sight (LOS) communication conditions and energy transmission efficiency, as shown in Fig. 2.

B. Nonlinear Energy Harvesting Model

In reflective A-IoT systems, devices contain small energy storage units (such as supercapacitors). Devices do not have an active radio frequency transmission capability, and their communication behavior is entirely based on reflecting CW carrier signals in the environment. Therefore, device energy is used mainly for the carrier reflection modulation process. Energy consumption is determined by reader-allocated power, specifically coupled with physical factors such as path loss and reflection efficiency. Devices harvest energy only when they are in an inactive state.

To realistically reflect the nonlinear characteristics of energy harvesting circuits, this paper adopts a nonlinear energy harvesting model based on the Sigmoid function. Traditional linear energy harvesting models cannot truly reflect the saturation characteristics of energy harvesting circuits, while nonlinear energy harvesting models can more accurately describe the process of devices that harvest energy from the environment [12], [13].

The actual energy harvesting power $P_{har}(n,t)$ of the n-th device in time interval t can be expressed as:

$$P_{har}(n,t) = \frac{M}{1 + \exp(-s(P_{in}(n,t) - z))} \tag{3}$$

where M is the saturation power threshold of the rectifier, that is, the maximum power harvested when the RF to DC circuit is driven to saturation, s controls the steepness of the nonlinear charging curve and z sets the turn-on point of the energy harvester and $P_{in}(n,t)$ denotes the actual RF power received by the device. In practice, the parameters M, s, and z can be obtained by curve fitting the sigmoid model to the measured input-output characteristics of the chosen energy harvesting.

Device energy consumption mainly originates from circuit control losses during the reflection modulation process, which we model as:

$$P_{con}(n,t) = \alpha \cdot g_n \cdot p_n(t) \tag{4}$$

where α is the backscatter loss coefficient of the device, g_n corresponds to the path loss of the link from the reader to the

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n-th device, and $p_n(t)$ corresponds to the transmission power assigned to the n-th device in time slot t.

In summary, to simulate energy harvesting changes for each device across continuous time slots, we adopt the following energy update model:

$$E_n(t+1) = \min\big\{E_n(t) + (1-\eta_n(t))P_{har}(n,t)T - \eta_n(t)P_{con}(n,t)T,\ E_{max}\big\} \tag{5}$$

where $E_n(t)$ is the energy harvested by the device at the beginning of time slot t, E_{max} is the device's maximum storage capacity, $\eta_n(t)$ indicates whether the device is activated, $P_{con}(n,t)$ is the equivalent power required for device reflection communication in the current time slot, T is the duration of each time slot, and $P_{har}(n,t)$ is the energy collected by the device through environmental RF signals in time slot t while in idle state.

C. Optimization Problem Formulation

In this work, $\eta_n(t)$ denotes the activation state of device n in the time interval t, defined as a binary variable ($\eta_n(t) = 1$ if active, $\eta_n(t) = 0$ if idle). The variable $p_n(t)$ represents the transmission power allocated to the device n.

The objective of this research is to jointly optimize the activation states of the device η and the power allocation p in the ambient IoT system under the constraints of energy causality and quality of service (QoS) to maximize system performance. The throughput of a single device is defined as:

$$R_n(t) = \eta_n(t) \cdot \log_2(1 + \text{SINR}_n(t)) \tag{6}$$

where $\eta_n(t)$ represents the activation state of the device n in time interval t, a binary variable (1 for active, 0 for idle); $\text{SINR}_n(t)$ represents the signal-to-interference plus noise ratio of device n in time interval t, specifically defined considering that device communication behavior depends on environmental RF signals:

$$\text{SINR}_n(t) = \frac{g_n^2 \cdot p_n(t)}{I_n(t) + \sigma^2} \tag{7}$$

where g_n corresponds to the loss of path of the link from reader to device n,the effective channel amplitude in the SINR expression appears as g_n^2, capturing the cascaded nature of the link, $p_n(t)$ corresponds to the transmission power of Device n in time slot t, and σ^2 is the noise power at the receiver.

Since devices in the considered A-IoT system have no active transmission capability and communicate solely by reflecting the continuous wave (CW) signals emitted by the reader, the effective link between reader and device is a cascaded backscatter channel. As a result, the signal experiences path attenuation twice (reader-to-device and device-to-reader) in addition to the device's reflection loss. This propagation characteristic is explicitly captured in the formulation of $\text{SINR}_n(t)$ above, making the model closely approximate actual A-IoT deployment scenarios.

Thus, the total throughput can be defined as follows:

$$R_{total} = \sum_{t=1}^{T}\sum_{n=1}^{N} R_n(t) \tag{8}$$

Since devices are passive and can only harvest energy in inactive states, performing energy updates during the optimization process would make the optimization problem extremely complex. Therefore, to simplify the optimization problem, this paper updates each device's energy harvesting state according to the optimization results and formula (3) after each time slot's optimization ends, ensuring the following energy causality constraints are satisfied:

$$\sum_{\tau=1}^{t} \eta_n(\tau)P_{con}(\tau)T \le E_n(0) + \sum_{\tau=1}^{t-1}(1-\eta_n(\tau))P_{har}(\tau)T \tag{9}$$

QoS constraints are expressed as the device's single time slot rate cannot be lower than the minimum required rate R_{min}:

$$\eta_n(t) \cdot R_n(t) \ge \eta_n(t) \cdot R_{min}, \quad \forall n, t \tag{10}$$

Furthermore, to ensure communication quality, the maximum number of access devices K is also specified:

$$\sum_{n=1}^{N} \eta_n(t) \le K \tag{11}$$

Finally, the optimization problem can be formally described as follows.

$$\begin{aligned} \max_{\eta,p} \quad & R_{total} \\ \text{s.t.} \quad & \text{Constraints (9), (10), (11)} \\ & 0 \le p_n(t) \le P_{max}, \quad \forall n, t \\ & \eta_n(t) \in \{0,1\}, \quad \forall n, t \end{aligned} \tag{12}$$

This optimization problem is a typical MINLP problem, challenging to solve, especially in practical industrial scenarios with numerous devices and complex constraints. Therefore, this paper designs efficient algorithms to solve it.

III. Optimization Algorithm Design

The joint optimization of device activation and power allocation formulated in (12) is inherently nonconvexx and combinatorial, posing significant challenges for conventional algorithms. This section first introduces the classical Lagrange dual method as a baseline, followed by a novel adaptive bilevel decomposition with branch-and-bound (ABD-BB) algorithm. We detail the motivation, methodology, and computational complexity analysis of the proposed algorithm.

A. Lagrange Dual Method

The Lagrange dual method is a classical approach effective for nonconvex optimization with complex constraints. By introducing dual variables, it transforms the original optimization problem into a more tractable dual formulation, solving primal and dual problems alternately.

we introduce dual multipliers λ, μ, and ν associated with constraints (9)-(11), respectively. The Lagrangian function is thus defined as

$$\mathcal{L}(\eta, p, \lambda, \mu, \nu) = R_{total} + \sum \lambda_i g_i + \sum \mu_j h_j + \sum \nu_k f_k \tag{13}$$

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Then, the dual optimization problem can be expressed as:

$$\min_{\lambda\geq 0,\mu\geq 0,\nu} \max_{\eta,p} \mathcal{L}(\eta, p, \lambda, \mu, \nu) \tag{14}$$

This dual problem can be solved iteratively by alternating between optimizing the primal variables (η, p) and updating the dual variables (λ, μ, ν). The subsequent subsections provide detailed procedures for solving these two subproblems.

1) Solving Power Allocation Variables: Taking the partial derivative of $\mathcal{L}$ with respect to $p_n(t)$ and setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial p_n(t)} = 0 \tag{15}$$

The partial derivative structure of (7) is as follows:

$$\frac{\partial R_n(t)}{\partial p_n(t)} = \frac{\eta_n(t) g_n^2}{\ln 2 \cdot (\sigma^2 + g_n^2 p_n(t))} \tag{16}$$

This yields the optimal power allocation:

$$p_n^{(t)} = \left[\frac{\eta_n(t)}{\ln 2 \cdot (\lambda_n^{(t)} + \mu_n^{(t)})} - \frac{\sigma^2}{g_n^2} \right]^+ \tag{17}$$

where $[x]^+ = \max(0, x)$ ensures nonnegative power.

2) Device Activation Strategy: For binary activation variables $\eta_n(t)$, the devices are activated based on marginal revenue:

$$\Delta_n(t) = \frac{\partial \mathcal{L}}{\partial \eta_n(t)} \tag{18}$$

The optimal strategy selects the top$K$ devices with the highest marginal revenue:

$$\eta_n(t) = \begin{cases} 1, & \text{if } n \in \text{Top-}K(\Delta(t)) \\ 0, & \text{otherwise} \end{cases} \tag{19}$$

3) Dual Variable Update: The dual variables are updated using the subgradient method to ensure the satisfaction of the constraint. The subgradients are as follows.

$$\begin{aligned} \nabla_{\lambda_i} \mathcal{L} &= g_i(\eta, p) \\ \nabla_{\mu_j} \mathcal{L} &= h_j(\eta, p) \\ \nabla_{\nu_k} \mathcal{L} &= f_k(\eta, p) \end{aligned} \tag{20}$$

The update rules are as follows:

$$\lambda_i^{(k+1)} = [\lambda_i^{(k)} - \alpha^{(k)} \nabla_{\lambda_i} \mathcal{L}]^+ \tag{21}$$

$$\mu_j^{(k+1)} = [\mu_j^{(k)} - \alpha^{(k)} \nabla_{\mu_j} \mathcal{L}]^+ \tag{22}$$

$$\nu_k^{(k+1)} = \nu_k^{(k)} - \alpha^{(k)} \nabla_{\nu_k} \mathcal{L} \tag{23}$$

where $\alpha^{(k)}$ is the step size and $[\cdot]^+$ maintains non-negativity.

B. Two-Layer Approximation with Adaptive Branch-and-Bound (ABD-BB) Algorithm

To tackle the non-convex and combinatorial nature of the throughput maximization problem, we develop a two-layer approximation with adaptive branch-and-bound (ABD-BB). The outer layer uses continuous relaxation with entropy regularization to score devices and shrink the binary search space, while the inner layer applies an adaptive branch-and-bound scheme to search this reduced space efficiently. Compared with Benders-type decompositions, which often yield weak cuts or require many iterations for non-convex MINLP, this two-layer design provides a more favourable trade-off between solution quality and computational cost.

The ABD-BB algorithm employs a hierarchical optimization structure: the outer layer uses continuous relaxation to rapidly identify promising device candidates, while the inner layer performs adaptive branch-and-bound to determine optimal activation patterns and power allocations. This two-layer approach effectively balances solution quality with computational efficiency, making it suitable for real-time industrial applications.

1) Outer Layer Continuous Relaxation and High-Potential Device Screening: The outer layer transforms the integer programming problem into a tractable continuous optimization. We relax binary activation variables $\eta_n(t) \in \{0, 1\}$ to continuous variables $\tilde{\eta}_n(t) \in [0, 1]$ and introduce regularization of entropy to promote sparsity.

$$\max_{\tilde{\eta},p} \sum_{t,n} \tilde{\eta}_n(t) \log_2(1 + \text{SINR}_n(t)) - \theta \sum_{t,n} \tilde{\eta}_n(t) \log \tilde{\eta}_n(t) \tag{24}$$

where θ is the sparsity regularization coefficient. The entropy term $-\tilde{\eta}_n(t) \log \tilde{\eta}_n(t)$ encourages solutions to converge toward binary values, facilitating the subsequent discretization process.

After solving this relaxed problem to obtain $\tilde{\eta}^*$, we perform the energy feasibility check. Devices with insufficient energy for activation are eliminated by setting $\tilde{\eta}_n(t) = 0$ if $E_n(t) < E_{threshold}$. From the remaining devices, we select the top$M$ scoring ones to form the candidate pool $\mathcal{C}$, where $M = \min(2K, N)$ ensures sufficient diversity while maintaining computational tractability.

2) Inner Layer Adaptive Combination Generation and Power Allocation: The inner layer explores the reduced search space defined by the candidate pool $\mathcal{C}$ to find the optimal combination of device activation. We generate L different activation patterns using a hybrid strategy:

First, a greedy combination is created by selecting the top-K devices from $\mathcal{C}$ based on their relaxation scores. This ensures at least one high-quality solution. Subsequently, $(L-1)$ additional combinations are generated through random sampling, where each combination randomly selects K devices from $\mathcal{C}$. This randomization helps escape local optima and explores diverse regions of the solution space.

For each activation pattern $\eta(l) \in \mathcal{H}$, where $\mathcal{H}$ denotes the set of all L combinations, we solve the power allocation subproblem:

$$\begin{aligned} \max_{p} \quad & \sum_{t,n} \eta_n(l,t) \log_2(1 + \text{SINR}_n(t)) \\ \text{s.t.} \quad & \text{Constraints (9), (10), (11)} \\ & 0 \leq p_n(t) \leq P_{max}, \quad \forall n, t \\ & \eta_n(t) \in \{0, 1\}, \quad \forall n, t \end{aligned} \tag{25}$$

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Algorithm 1 ABD-BB Algorithm

Input: System parameters N, K, T, $E_n(0)$, channel gains g_n
Output: Optimal η^*, p^* and throughput R_{total}

Initialize: $\theta \leftarrow 0.1$, $M \leftarrow \min(2K, N)$, $L \leftarrow 10$
for $t = 1$ to T **do**
  Outer Layer:
  Solve relaxed problem (15) to obtain $\tilde{\eta}^*$
  Energy Feasibility Screening:
  for $n = 1$ to N **do**
    if $E_n(t) < E_{threshold}$ **then**
      $\tilde{\eta}_n(t) \leftarrow 0$
    end if
  end for
  Select top-M devices by score $\rightarrow$ candidate pool $\mathcal{C}$
  Inner Layer:
  $\eta(1) \leftarrow$ Top-K devices from $\mathcal{C}$
  for $l = 2$ to L **do**
    $\eta(l) \leftarrow$ Random K devices from $\mathcal{C}$
  end for
  Power Allocation:
  for $l = 1$ to L **do**
    Solve subproblem (16) for $\eta(l) \rightarrow p(l)$
    Calculate $R_{total}(l)$
  end for
  $(\eta_t^*, p_t^*) \leftarrow \arg\max_l R_{total}(l)$
  Update energy states using (5)
end for
return η^*, p^*, R_{total}

This subproblem, with fixed binary variables, becomes a convex optimization problem that can be efficiently solved using augmented Lagrangian methods or interior-point algorithms. The solution ensures strict satisfaction of all constraints while maximizing throughput for the given device activation pattern.

After evaluating all combinations, we select the one that yields maximum system throughput:

$$(\eta^*, p^*) = \arg \max_{(\eta(l), p(l))} R_{total}(\eta(l), p(l)) \tag{26}$$

3) Algorithm Flow and Complexity Analysis: Algorithm 1 presents the complete ABD-BB procedure. The algorithm processes each time slot sequentially, maintaining the continuity of the energy state across the slots. The computational complexity of ABD-BB consists of two main components. The outer layer requires $O(N^3)$ operations to solve the continuous relaxation problem using standard convex optimization solvers. The inner layer needs $O(LN^2)$ operations, where each of the L power allocation sub-problems requires $O(N^2)$ computations. Therefore, the overall complexity is $O(N^3 + LN^2)$ per time slot, which is polynomial in system size.

Compared to exhaustive branch-and-bound with exponential complexity $O(2^N)$, ABD-BB achieves dramatic computational savings. For a system with $N = 18$ devices, traditional methods require evaluating $2^{18} = 262,144$ combinations, while ABD-BB examines only carefully selected combinations $L = 10$.

TABLE I
SIMULATION PARAMETER SETTINGS

Parameter Name	Symbol	Value/Unit
Total number of devices	N	18
Maximum activated devices	K	2
Number of time slots	T	100
Maximum transmission power	P_{max}	0.2 W
Minimum rate requirement	R_{min}	0.5 bit/s/Hz
Noise power	σ^2	10^{-10} W
Time slot length	T_{slot}	1 ms
Initial energy	$E_n^{(0)}$	10^{-3} J
Maximum energy capacity	E_{max}	10^{-3} J
Energy harvesting parameters	M, s, z	0.24, 150, 0.014
Communication frequency	f	837 MHz
Relaxation regularization	θ	0.1
Sparsity penalty coefficient	λ	0.01 $\sim$ 0.05

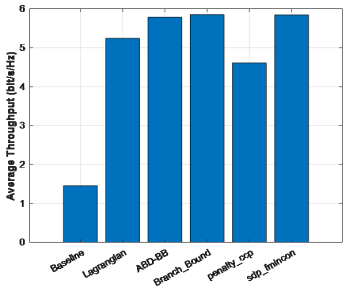


Fig. 3. Comparison of average throughput across different algorithms.

This efficiency gain enables real-time optimization in industrial IoT systems where rapid adaptation to changing conditions is essential.

IV. SIMULATION RESULTS

In this section, simulation results are presented to validate the effectiveness and superiority of the proposed ABD-BB algorithm compared to existing methods. The simulation setup parameters are summarized in Table I, with reference to the industrial indoor environment parameters provided by 3GPP TR 38.769.

Fig. 3 compares the average throughput of six algorithms. The proposed ABD-BB method achieves approximately 5.78 bits/s/Hz, on par with the near-optimal Branch&Bound and an SDP-based solution (5.85 and 5.94 bits/s/Hz, respectively), and far above the baseline 1.45 bits/s/Hz. It also outperforms other heuristics (Lagrangian and penalty-CCP), demonstrating the effectiveness of our approach.

Fig. 4 shows the per-slot throughput over time. The proposed algorithm and the near-optimal methods (Branch-Bound, SDP) sustain a consistently high throughput of around 6 bits/s/Hz in each time slot, with minimal fluctuation after initial convergence. In contrast, the baseline's throughput varies widely between 0 and 3 bits/s/Hz (even dropping to 0 in some slots), indicating unreliable performance. These results highlight the superior stability and reliability of our approach.

Fig. 5 and Fig. 6 analyze the influence of the regularization coefficient θ and candidate pool size on performance and

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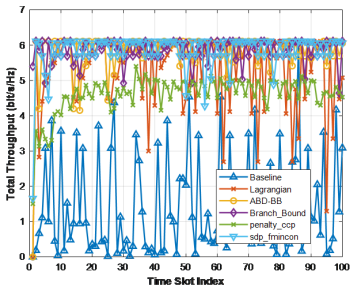


Fig. 4. Throughput variation at each time slot.

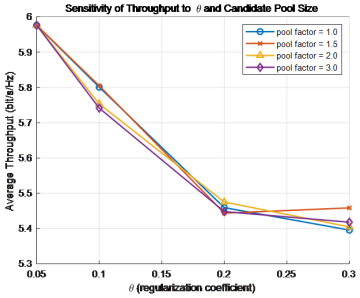


Fig. 5. Impact of regularization coefficient θ on average throughput.

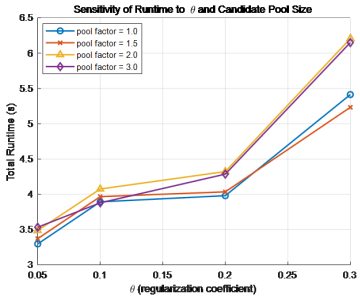


Fig. 6. Impact of candidate pool size on total runtime.

complexity. In Fig. 5, the average throughput slightly decreases as θ increases (dropping from about 5.98 to 5.43 bits/s/Hz when θ rises from 0.05 to 0.2), indicating that an overly large θ imposes excessive regularization and reduces performance. Fig. 6 shows that the total runtime grows with larger θ and pool size: increasing θ from 0.05 to 0.3 nearly doubles the runtime (from 3.5 s to 6.0 s), and using a bigger pool factor (3.0 vs 1.0) adds about 0.71 s. These results reveal a trade-off, as moderate θ and pool choices achieve high throughput with reasonable runtime, whereas excessive values lead to diminishing returns and higher complexity.

V. Conclusion

In this paper, we proposed the ABD-BB algorithm for joint device activation and power allocation in A-IoT systems under RF energy harvesting constraints. Simulation results demonstrate that ABD-BB achieves an average throughput of 5.78 bits/s/Hz, representing a 297.5% improvement over the random baseline (1.45 bits/s/Hz). Compared with other heuristics, such as the Lagrangian (5.24 bits/s/Hz, 260.1%) and penalty-CCP (4.61 bits/s/Hz, 216.9%), ABD-BB provides clear gains, while approaching the near-optimal Branch-and-Bound (5.85 bits/s/Hz, 302.1%) and SDP (5.94 bits/s/Hz, 308.0%) solutions with only marginal gaps. These results confirm that ABD-BB delivers near-optimal throughput with substantially lower complexity than exhaustive methods. Moreover, sensitivity analysis shows that proper choices of the regularization coefficient and candidate pool size balance performance and runtime, enabling real-time deployment in industrial environments. Future work will extend ABD-BB to higher device densities, dynamic scheduling scenarios, and multi-objective optimization considering energy efficiency and fairness.

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